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Semantic-Guided Visual Representation Construction for Remote Sensing Image–Text Retrieval

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***Abstract*—Remote sensing image–text retrieval (RSITR) continues to face significant challenges in achieving precise cross-modal semantic correspondence. While CLIP-based methods are effective at instance-level alignment, most of them construct visual representations independently of individual text queries and rely on pairwise contrastive supervision. Consequently, fixed image embeddings may mix query-relevant evidence with visual content unmentioned by the query, while the shared embedding space may exhibit geometric inconsistency. Such geometric inconsistency manifests as inconsistent neighborhood relationships at the local level and persistent directional discrepancies at the global level across modalities. To address these problems, we propose a Semantic-Guided Geometry-Aware Alignment (SGGA) framework for RSITR, centered on query-conditioned visual representation construction and further enhanced by local and global geometric regularization. Central to SGGA is a Semantic-Guided Patch Aggregation (SGPA) module that uses the text embedding as a semantic query to dynamically aggregate patch-level visual features, producing a candidate-specific visual representation for each image–text pair. To further regulate the geometry of the resulting query-conditioned embedding space, we design Bidirectional Neighborhood Distribution Alignment (BNDA) and Shared Anchor Direction Regularization (SADR) to preserve local neighborhood consistency and reduce global directional discrepancy across modalities, respectively. Furthermore, we design a gap-aware dual-query inference strategy that constructs complementary SGPA queries from the original sentence embedding and a gap-aware text centroid, and fuses their similarity scores to improve retrieval robustness. Experiments on three RSITR datasets demonstrate that SGGA achieves state-of-the-art (SOTA) retrieval performance.**

***Index Terms*—Remote sensing image-text retrieval, vision-language alignment, semantic-guided representation, geometric consistency.**

I. Introduction

REMOTE sensing image–text retrieval [1] aims to establish bidirectional semantic correspondence between remote sensing imagery and natural language descriptions, and plays a crucial role in applications such as resource surveying [2], [3], disaster monitoring [4], [5], and geographic intelligence analysis [6]–[8]. Despite recent progress, accurate cross-modal matching in remote sensing remains challenging due to the intrinsic heterogeneity between visual and textual modalities. Remote sensing images typically contain multi-scale objects, complex background clutter, and dense spatial layouts [9], while textual descriptions [2], [10] are often concise and cover only a subset of salient semantic elements. This semantic asymmetry hinders precise cross-modal alignment and limits retrieval performance.

In recent years, Contrastive Language–Image Pre-training (CLIP) [11] has become a dominant paradigm for RSITR [12]–[14]. By projecting visual and textual representations into a shared semantic embedding space, CLIP provides a strong semantic prior for associating remote sensing visual patterns with natural-language descriptions. However, existing methods exhibit two limitations. Remote sensing images often contain richer scene content than that described by a concise text query, while their visual representations are typically constructed independently of the query. A fixed image embedding may therefore mix query-relevant evidence with unmentioned visual content, weakening query-specific similarity estimation, as illustrated in Fig. 1(a) and quantitatively examined in Section V-B3. Furthermore, pairwise contrastive supervision does not explicitly regulate the local and global geometry of the shared embedding space, as shown in Fig. 1(c).

Several studies have adapted CLIP to RSITR through parameter-efficient tuning, prior-guided representation refinement, and multi-scale semantic alignment [15]–[20]. For example, Cross-Modal Adapter [21] introduces lightweight adaptation modules into the visual and textual encoders, PIR [22] exploits prior knowledge to reduce semantic noise, and MSSA [23] enhances multi-scale semantic correspondence. Despite their effectiveness, these methods generally retain a query-independent global image representation for final retrieval, even when textual information is introduced during feature refinement, auxiliary interaction, or similarity modeling. Consequently, the same image embedding is applied to descriptions emphasizing different objects or spatial relations, which may limit query-specific discrimination. This limitation motivates candidate-specific visual representation construction conditioned on the semantics of each text query.

Beyond instance-level correspondence, reliable cross-modal retrieval also depends on the geometric consistency of the shared embedding space. Pairwise contrastive supervision primarily optimizes the discrimination of matched and mismatched pairs, without explicitly preserving inter-sample relationships or regulating modality-level feature directions. Consequently, semantically related samples may exhibit inconsistent neighborhood relationships across modalities at the local level [24]–[26], while visual and textual representations

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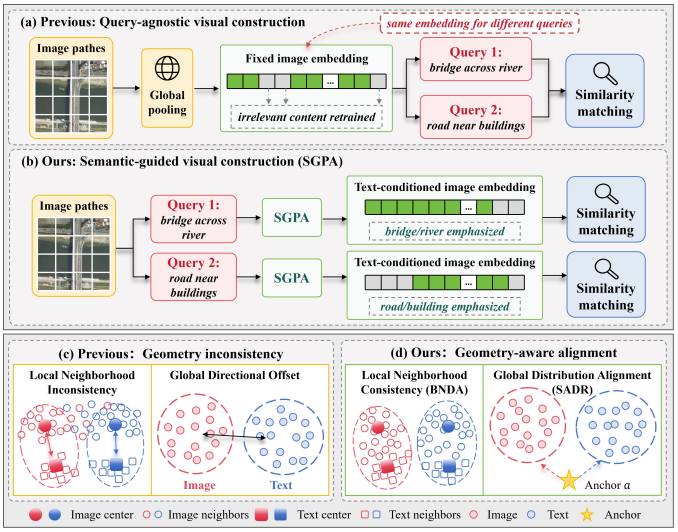


Fig. 1. Motivation for the SGGA framework.(a) Conventional methods construct fixed image embeddings independently of individual text queries, which may mix query-relevant evidence with unmentioned visual content. (b) SGPA guides visual aggregation with each text query to generate query-conditioned image representations. (c) Contrastive alignment does not explicitly regulate local neighborhood consistency or global directional discrepancy. (d) BNDA preserves local neighborhood consistency, while SADR reduces global directional discrepancy through shared anchor-direction regularization.

may retain persistent directional discrepancy at the global level [27], [28]. Although existing RSITR methods improve pairwise alignment or specific structural relationships, joint regulation of these local and global geometric inconsistencies remains limited [24], [29]–[31], potentially reducing the reliability of cross-modal similarity estimation.

To address these limitations, we propose SGGA, a unified framework that couples query-conditioned visual representation construction with local and global geometric regularization. SGPA constructs a candidate-specific visual embedding for every image–text pair in the complete similarity matrix, as illustrated in Fig. 1(b). Specifically, the text embedding guides the aggregation of patch-level visual tokens, enabling the representation of the same image to adapt to descriptions emphasizing different objects, attributes, or spatial relations. The resulting query-conditioned embedding space is further regulated by BNDA and SADR through complementary geometric constraints. BNDA independently constructs text-defined and visual-defined neighborhoods and aligns their similarity distributions in both directions, preserving local neighborhood relationships over complementary neighborhood supports. SADR regularizes the visual and textual modality centers toward an EMA-stabilized shared anchor direction, reducing modality-level directional discrepancy while retaining the instance-level discrimination learned by contrastive supervision, as illustrated in Fig. 1(d). BNDA and SADR are applied to the positive-pair SGPA-conditioned visual representations and are jointly optimized with the contrastive objective, forming a coherent representation-and-geometry alignment framework. During inference, SGGA derives a complementary text centroid from a modality-gap direction estimated on the validation set and fuses its SGPA similarity score with that obtained from the original sentence query to improve retrieval robustness.

The main contributions of this work are as follows.

1) We propose SGGA, a unified RSITR framework that enables textual semantics to directly condition visual representation construction. SGPA generates a candidate-specific visual embedding for each image–text pair, allowing the same image to be represented according to the semantic focus of different descriptions.
2) We introduce complementary geometric constraints for the resulting query-conditioned embedding space. BNDA aligns similarity distributions over separately constructed text-defined and visual-defined neighborhoods to preserve local neighborhood relationships, while SADR reduces modality-level directional discrepancy through an EMA-stabilized shared anchor.
3) We design a gap-aware dual-query retrieval strategy that fuses SGPA similarity scores from the sentence embedding and a gap-aware text centroid to improve retrieval robustness.
4) We conduct extensive experiments on RSICD, RSITMD, and UCM-Captions, where SGGA achieves the best overall mR among the compared methods and shows consistent cross-dataset transfer improvements.

II. RELATED WORK

A. CLIP-based RSITR

CLIP has become the dominant paradigm for RSITR by projecting visual and textual features into a shared semantic embedding space through contrastive learning. Its strong generalization ability and transferable representations make it particularly appealing for remote sensing applications, where annotated image–text pairs are relatively scarce and scene semantics are often complex [2]. Building upon CLIP, a series of methods [12], [13], [32], [14] have been proposed to adapt it to the remote sensing domain and improve cross-modal semantic correspondence.

To improve cross-modal semantic correspondence in remote sensing scenes, existing methods have explored several directions. RemoteCLIP [12] constructs a curated remote sensing image–text dataset for domain-adaptive pre-training, while GeoRSCLIP [13] further scales this to millions of pairs, both significantly strengthening the semantic prior for remote sensing scene understanding. Beyond above retrieval-oriented models, GeoChat [33] develops a grounded generative VLM for remote sensing instruction following and region-level dialogue. To better adapt CLIP to the unique visual characteristics of remote sensing imagery, CLIP-Adapter [32] and Cross-Modal Adapter [21] introduce lightweight adaptation modules that refine pretrained representations without large-scale retraining, while soft-label guided multi-granularity prompt learning further demonstrates the effectiveness of combining visual and textual semantic guidance [34]. PE-RSITR [14] further adopts parameter-efficient transfer learning to reduce the domain gap with limited additional cost. Recognizing that remote sensing images contain dense spatial layouts and multi-scale objects that are often overlooked by global feature

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matching, MSSA [23] incorporates multi-scale feature aggregation to improve sensitivity to local semantic cues. Along this line, region–word [35] and patch–token alignment [36] strategies are introduced to establish finer correspondences between image regions and textual descriptions, reducing the interference of irrelevant visual content during matching. Furthermore, knowledge-aware methods [8] leverage external semantic information to handle concise and ambiguous text queries, while dark-knowledge [37] association guided hashing exploits implicit CLIP knowledge and cross-modal relations for unsupervised cross-modal retrieval, enabling more discriminative semantic correspondence.

Despite these advances, most existing methods share a common limitation in how visual representations are constructed: textual information is introduced only at the matching or projection stage, rather than directly guiding visual representation construction. As a result, fixed image embeddings may mix query-relevant evidence with visual content unmentioned by the query, and the semantic asymmetry between complex remote sensing scenes and concise text queries further amplifies this interference. Text-conditioned cross-attention has also been explored in the broader vision-language literature [36] for single-pair token-level reranking, with cost growing with the number of candidates. To overcome this limitation, we propose the SGPA module, which uses the text embedding as a semantic query to dynamically aggregate patch-level visual features. This enables query-relevant visual content to be selectively emphasized during visual representation construction, effectively suppressing irrelevant visual regions in complex remote sensing scenes.

B. Geometric Consistency in Cross-Modal Embedding Space

Although CLIP provides a unified embedding space for cross-modal matching, the shared embedding space learned by contrastive objectives is not geometrically consistent across modalities [24], [38]. Visual and textual features tend to occupy separated sub-regions of the shared space, and pairwise contrastive supervision alone does not explicitly regulate the geometric structure of the embedding space, leaving inconsistent neighborhood relationships and persistent directional discrepancy unaddressed [28], [39]. These geometric inconsistencies undermine the reliability of cross-modal similarity estimation, particularly in remote sensing scenes where semantically similar image–text pairs exhibit high intra-modal variation.

To improve the geometric consistency of the shared embedding space, existing methods have explored two directions. At the local level, topology- and structure-aware methods introduce explicit structural constraints to preserve relationships among semantically related features, with structure-aware Transformer [40] also demonstrating the value of adaptive structural modeling in visual representation learning. Representative works incorporate pairwise similarity preservation [25], relative order constraints [41], or graph-based regularization [42] to maintain consistent neighborhood relationships during cross-modal mapping, thereby improving local retrieval ranking. At the global level, distribution alignment methods attempt to reduce the overall discrepancy between image and text feature distributions. Methods such as AlignCLIP [27] and distribution-regularized alignment frameworks [43], [44] have been proposed to address global geometric inconsistency. These methods apply feature normalization [45], or reference-based alignment [46] to improve the global compactness and distributional consistency of the embedding space.

However, existing methods exhibit limitations, leaving the geometric consistency of the shared embedding space only partially addressed. At the local level, topology-preserving methods rely on pairwise similarity or ranking-based constraints, which cannot explicitly enforce bidirectional consistency between image-to-text and text-to-image neighborhood distributions. At the global level, distribution alignment methods lack explicit directional regularization, leaving persistent directional discrepancy between modalities unaddressed. To overcome these limitations, we propose BNDA and SADR. BNDA recalibrates visual similarity with text-derived semantic structure and enforces bidirectional distribution alignment over stop-gradient top-k neighborhoods, improving local geometric consistency without relying on raw pairwise similarity. SADR aligns both modality centers toward an EMA-stabilized anchor jointly derived from their perturbed centers, reducing global directional discrepancy without biasing toward either modality.

III. METHOD

A. Overview of the SGGA Framework

To address the above challenges, we propose the SGGA framework, centered on SGPA for query-conditioned visual representation construction and further enhanced by BNDA and SADR for geometry-aware alignment of the shared embedding space. The overall architecture of SGGA is illustrated in Fig.2.

Given an image–text mini-batch, the visual encoder produces patch-level visual tokens and a global visual embedding, while the text encoder produces a global sentence embedding. SGPA uses the sentence embedding as a query to dynamically aggregate patch-level visual tokens via cross-attention, producing a query-conditioned visual representation that emphasizes query-relevant regions while suppressing background interference. On this basis, BNDA preserves local neighborhood relationships by aligning neighborhood-restricted similarity distributions across modalities through bidirectional KL divergence, ensuring that semantically related samples remain locally coherent in the shared embedding space. SADR reduces persistent directional discrepancy between modalities by aligning modality-level feature centers toward an EMA-stabilized shared reference direction.

The overall objective is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CLIP}}(\tilde{\mathbf{f}}^v, \mathbf{f}^t) + \lambda_B \mathcal{L}_{\text{BNDA}} + \lambda_S \mathcal{L}_{\text{SADR}}, \quad (1)$$

where $\tilde{\mathbf{f}}^v$ denotes the SGPA-conditioned visual embedding and $\mathbf{f}^t$ denotes the global text embedding. λ_B and λ_S control the contributions of BNDA and SADR, respectively.

During inference, SGPA generates query-conditioned visual representations under two complementary text-side queries: the global sentence embedding and a gap-aware text centroid.

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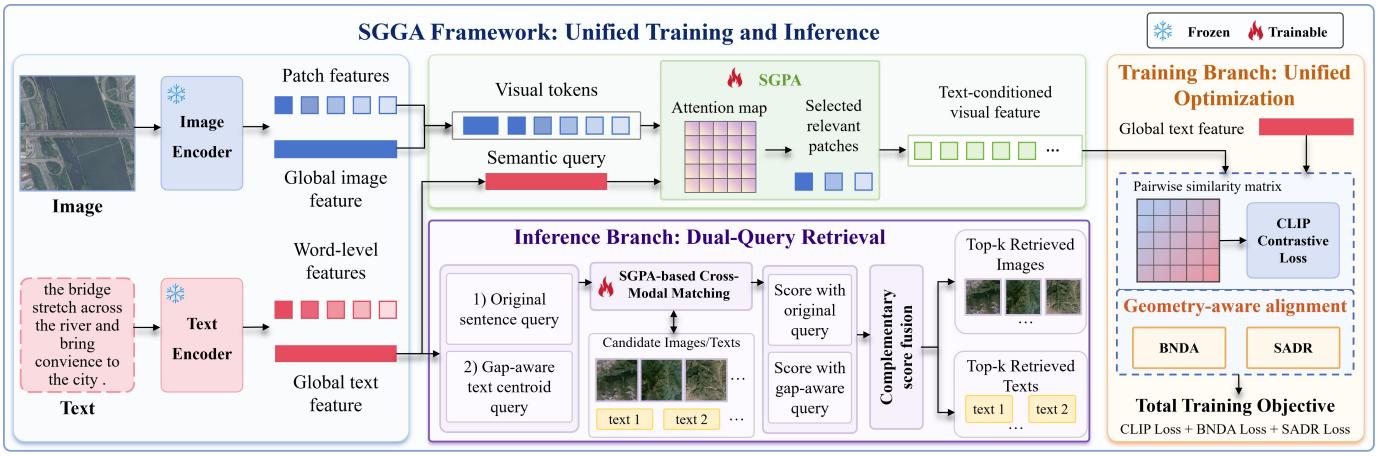


Fig. 2. Overview of the SGGA framework. SGPA serves as the shared core of SGGA, using the global text feature as a semantic query to construct query-conditioned visual representations from visual tokens. During training, the SGPA-conditioned space is optimized by CLIP contrastive alignment, BNDA-based local consistency, and SADR-based global consistency. During inference, two complementary text-side queries are used for SGPA-based matching and score fusion to obtain the final top-k retrieval results.

The corresponding similarity scores are adaptively fused to account for residual cross-modal discrepancy and produce the final retrieval ranking.

B. Semantic-Guided Patch Aggregation

In RSITR, visual representations used for retrieval are typically constructed without explicit guidance from the text query, causing fixed image embeddings to mix query-relevant evidence with visual content unmentioned by the query. To address this, we propose the SGPA module, as illustrated in Fig. 3. SGPA introduces the text embedding as a query signal to dynamically aggregate patch-level visual features via multi-head cross-attention, enabling the visual representation to adapt to different textual semantics. Since the candidate text directly conditions visual aggregation, SGPA can be characterized as a lightweight query-conditioned reranker rather than a query-independent two-tower encoder. The aggregated feature is then fused with the original global representation through a gated residual connection, which enhances query-relevant information while preserving overall representational stability.

Given the patch-level visual features $F_{\text{patch}} \in \mathbb{R}^{N\times D}$ and the global image feature $f_{\text{img}} \in \mathbb{R}^{D}$, we construct an augmented visual token sequence by prepending the global image feature as an additional token:

$$\hat{F} = [f_{\text{img}};\, F_{\text{patch}}] \in \mathbb{R}^{(N+1)\times D}. \tag{2}$$

In this way, the original global feature is retained as a complementary token rather than serving as the sole visual representation, allowing the cross-attention mechanism to jointly attend over both patch-level details and the holistic image context.

The query is projected from the text embedding, while the keys and values are projected from the augmented visual token sequence:

$$Q = \phi_Q(f_{\text{txt}}), \quad K = \phi_K(\hat{F}), \quad V = \phi_V(\hat{F}), \tag{3}$$

Here, ϕ_Q and ϕ_K consist of a LayerNorm layer followed by a bottleneck MLP with a Linear down layer, GELU activation, and a Linear up layer with intermediate dimension $D/2$, while ϕ_V is a linear projection. This asymmetric design applies stronger nonlinear transformation to the query and key projections to capture semantic alignment, while keeping the value projection lightweight to preserve feature fidelity.

The attention computation is described for a single image–text pair, with the batch dimension omitted. After projection, the query, key, and value are reshaped into H heads with per-head dimension $d_h = D/H$. For the h-th head, the text query is $q_h \in \mathbb{R}^{d_h}$, and the visual keys and values are $K_h \in \mathbb{R}^{(N+1)\times d_h}$ and $V_h \in \mathbb{R}^{(N+1)\times d_h}$, respectively. The attention weight over the visual tokens is computed as:

$$a_h = \text{softmax}\left(\frac{K_h q_h}{\sqrt{d_h}}\right), \quad a_h \in \mathbb{R}^{N+1}, \tag{4}$$

The softmax is applied over all $(N+1)$ visual tokens, enabling the text query to selectively attend to query-relevant patches while suppressing irrelevant background regions. The output of the h-th head is then obtained by weighted aggregation:

$$\text{head}_h = a_h^{\top} V_h, \quad \text{head}_h \in \mathbb{R}^{d_h}. \tag{5}$$

The outputs of all heads are concatenated and projected to obtain a residual feature:

$$\delta = W_o \cdot \text{Concat}(\text{head}_1, \ldots, \text{head}_H), \quad \delta \in \mathbb{R}^{D}, \tag{6}$$

Here, $W_o \in \mathbb{R}^{D\times D}$ is the output projection matrix. The residual δ captures the query-conditioned visual content that complements the original global representation.

The residual is fused with the original global feature via a learnable gate:

$$\tilde{f}_{\text{img}} = \text{Norm}(f_{\text{img}} + \tanh(\alpha)\cdot\delta), \tag{7}$$

where $\text{Norm}(\cdot)$ denotes L2 normalization and α is a scalar gating parameter. The gating parameter α is initialized to a small value, such that the model initially relies on the original global feature and gradually incorporates the text-guided residual as training progresses, ensuring stable optimization from the start. To further stabilize early training, the output

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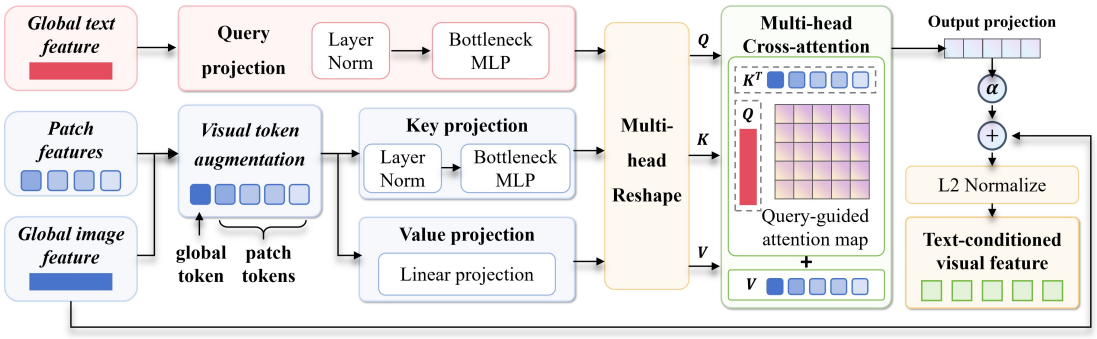


Fig. 3. Illustration of the SGPA module. The global text feature is used as the query, while the global image feature is prepended to patch features to form augmented visual tokens as keys and values. SGPA performs multi-head cross-attention and semantic-guided aggregation, then fuses the aggregated feature with the original global image feature through a learnable gated residual connection and L2 normalization to produce the query-conditioned visual representation.

projection W_o is initialized with a small variance, and the up-projection weights of ϕ_Q and ϕ_K are initialized to zero, so that the projected query and key features start near zero, yielding initially uniform attention weights across all visual tokens. As training progresses, the model gradually learns to concentrate attention on query-relevant regions.

The resulting $\tilde{f}_{\text{img}}$ serves as the query-conditioned visual representation for subsequent cross-modal similarity computation. During training, SGPA is applied in a vectorized manner over all image–text pairs in the mini-batch. For each text query, the corresponding query-conditioned image representations are computed simultaneously across all images to form the full similarity matrix for contrastive optimization. During inference, the same mechanism supports efficient construction of query-conditioned visual features under different text-side queries, which forms the basis of the dual-query retrieval strategy described in Section III-E.

C. Bidirectional Neighborhood Distribution Alignment

While SGPA enhances the discriminability of query-conditioned visual representations, contrastive learning alone does not guarantee that relative similarity relationships among samples are preserved across modalities, leading to inconsistent neighborhood relationships that degrade retrieval performance. To address this, we propose the BNDA module, which models each neighborhood as a modality-specific similarity distribution and enforces consistency between corresponding distributions across modalities, as illustrated in Fig. 4.

1) Intra-modal Similarity and Neighborhood Construction: Given a mini-batch of B image–text pairs, we obtain L2-normalized image and text embeddings $\{f_i^v\}_{i=1}^B$ and $\{f_i^t\}_{i=1}^B$, respectively. The intra-modal similarity matrices are computed as

$$S_{ij}^v = \langle f_i^v, f_j^v \rangle, \quad S_{ij}^t = \langle f_i^t, f_j^t \rangle, \tag{8}$$

where S^v and S^t describe the visual and textual neighborhood structures within the current mini-batch, respectively.

2) Semantic-Aware Similarity Rescaling: In remote sensing scenarios, visual similarity may be affected by background clutter, scale variations, and repetitive spatial patterns, making it less reliable as a direct indicator of semantic relatedness. For instance, two images may appear visually similar due to shared textures or land-cover patterns while describing different semantic contents. In contrast, textual embeddings are typically more compact at the semantic level and less affected by such visual ambiguities. Therefore, we use the semantic structure in the text space as a reference to recalibrate visual-side similarity logits.

The semantic distance is computed from a detached copy of the text similarity as

$$D_{ij} = 1 - S_{ij}^t. \tag{9}$$

To avoid scale imbalance across different anchors, we normalize the semantic distance within each row while excluding the anchor sample itself:

$$\bar{D}_{ij} = \frac{D_{ij}}{\frac{1}{B-1}\sum_{u \neq i} D_{iu} + \epsilon}, \quad i \neq j, \tag{10}$$

where ϵ is a small constant for numerical stability. The visual-side similarity logit is then recalibrated by subtracting the semantic distance penalty:

$$\tilde{S}_{ij}^v = S_{ij}^v - \gamma \bar{D}_{ij}, \quad i \neq j, \tag{11}$$

where $\gamma \geq 0$ controls the strength of semantic-aware rescaling. The text-derived semantic distance $\bar{D}_{ij}$ is used to rescale visual similarities before neighborhood construction, suppressing visually similar but semantically distant samples. When $\gamma = 0$, $\tilde{S}^v = S^v$ and semantic rescaling is disabled.

3) Masked Top-k Neighborhood Construction: For each anchor sample, we construct modality-specific neighborhoods according to the text-side similarity and the recalibrated visual-side similarity. Before top-k selection, the diagonal entries are masked out to exclude the anchor sample itself:

$$\bar{S}_{ij}^t = \begin{cases} S_{ij}^t, & i \neq j, \\ -\infty, & i = j, \end{cases} \quad \bar{S}_{ij}^v = \begin{cases} \tilde{S}_{ij}^v, & i \neq j, \\ -\infty, & i = j. \end{cases} \tag{12}$$

The top-k neighborhoods are then obtained as

$$\mathcal{N}_i^t = \text{TopK}(\bar{S}_{i,:}^t, k), \quad \mathcal{N}_i^v = \text{TopK}(\bar{S}_{i,:}^v, k). \tag{13}$$

The diagonal masking prevents the trivial self-neighbor from dominating the neighborhood distribution. The top-k indices are selected from stop-gradient copies of $\bar{S}^t$ and $\bar{S}^v$, while the corresponding differentiable similarity logits are retained for subsequent probability normalization and distribution alignment.

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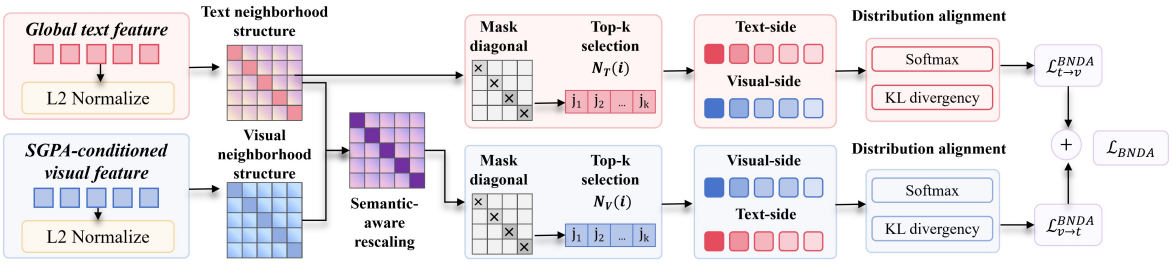


Fig. 4. Illustration of the BNDA module. BNDA constructs intra-modal similarity matrices from global text features and SGPA-conditioned visual features. Recalibrates visual similarity using text-derived semantic structure, and builds masked top-k neighborhoods. Bidirectional distribution alignment is performed over text-defined and visual-defined neighborhoods to preserve local neighborhood relationships.

4) Bidirectional Distribution Alignment: BNDA aligns the neighborhood distributions bidirectionally.For a selected support $\mathcal{N}_i \in \{\mathcal{N}_i^t, \mathcal{N}_i^v\}$, the modality-specific neighborhood distributions are defined by temperature-scaled softmax:

$$p_{ij}^{t,\mathcal{N}_i} = \frac{\exp(S_{ij}^t/\tau)}{\sum_{l\in\mathcal{N}_i}\exp(S_{il}^t/\tau)}, \quad j \in \mathcal{N}_i$$
$$p_{ij}^{v,\mathcal{N}_i} = \frac{\exp(\tilde{S}_{ij}^v/\tau)}{\sum_{l\in\mathcal{N}_i}\exp(\tilde{S}_{il}^v/\tau)}, \quad j \in \mathcal{N}_i. \tag{14}$$

In the text-to-image direction, the text-defined neighborhood $\mathcal{N}_i^t$ serves as the reference support, and the alignment loss is computed as

$$\mathcal{L}_{t\to v} = \frac{1}{B}\sum_{i=1}^{B}\mathrm{KL}\left(p_i^{t,\mathcal{N}_i^t} \| p_i^{v,\mathcal{N}_i^t}\right), \tag{15}$$

which encourages the visual similarity distribution to match the local neighborhood structure defined in the text space. In the image-to-text direction, the image-defined neighborhood $\mathcal{N}_i^v$ serves as the reference support:

$$\mathcal{L}_{v\to t} = \frac{1}{B}\sum_{i=1}^{B}\mathrm{KL}\left(p_i^{v,\mathcal{N}_i^v} \| p_i^{t,\mathcal{N}_i^v}\right), \tag{16}$$

which encourages the text-side similarity distribution to preserve the local structure induced by the recalibrated visual space. The final BNDA loss is formulated as

$$\mathcal{L}_{\mathrm{BNDA}} = \frac{1}{2}\left(\mathcal{L}_{t\to v} + \mathcal{L}_{v\to t}\right). \tag{17}$$

Thus, semantic-aware rescaling modifies the visual similarities before neighborhood construction, while the subsequent bidirectional KL alignment matches the normalized neighborhood distributions across modalities. The probability distributions are not detached, allowing the alignment losses to propagate gradients to both visual and textual embeddings.

D. Shared Anchor Direction Regularization

While BNDA preserves local neighborhood relationships, it does not constrain the global directional bias between visual and textual feature distributions, which can bias cross-modal similarity estimation. To mitigate this, we introduce Shared Anchor Direction Regularization (SADR), which regularizes the modality-level feature centers toward an EMA-stabilized shared reference direction, as illustrated in Fig. 5.

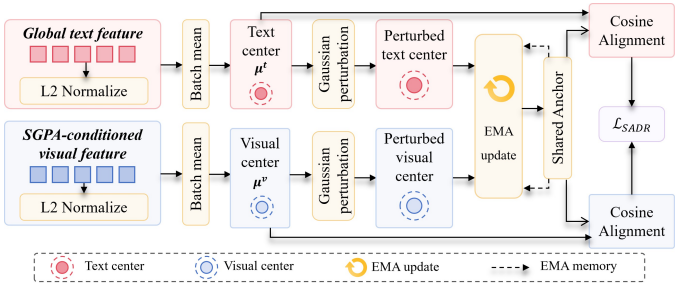


Fig. 5. Illustration of the SADR module. SADR constructs an EMA-stabilized shared anchor direction from textual and SGPA-conditioned visual centers with Gaussian perturbation. Both modality centers are regularized toward the shared anchor to reduce global directional discrepancy between modalities.

Given a mini-batch of B image–text pairs, let $\tilde{\mathbf{f}}_i^v$ denote the SGPA-conditioned visual embedding of the i-th image and $\mathbf{f}_i^t$ denote the corresponding global text embedding. The visual and textual modality centers within the current mini-batch are computed as

$$\boldsymbol{\mu}^v = \mathrm{Norm}\left(\frac{1}{B}\sum_{i=1}^{B}\tilde{\mathbf{f}}_i^v\right), \boldsymbol{\mu}^t = \mathrm{Norm}\left(\frac{1}{B}\sum_{i=1}^{B}\mathbf{f}_i^t\right), \tag{18}$$

where $\mathrm{Norm}(\cdot)$ denotes L2 normalization. Operating on modality-level centers rather than individual instances reduces global directional discrepancy while avoiding excessive interference with instance-level discriminability.

We first apply Gaussian perturbation to the modality centers to improve the robustness of the shared direction against mini-batch fluctuations:

$$\tilde{\boldsymbol{\mu}}^v = \mathrm{Norm}(\boldsymbol{\mu}^v + \epsilon^v), \quad \tilde{\boldsymbol{\mu}}^t = \mathrm{Norm}(\boldsymbol{\mu}^t + \epsilon^t), \tag{19}$$

where $\epsilon^v, \epsilon^t \sim \mathcal{N}(0, \sigma^2)$ are small Gaussian noise terms injected. Specifically, $\epsilon^v, \epsilon^t \in \mathbb{R}^{512}$ are independently resampled at each training step with i.i.d. per-dimension noise, $\sigma = 1.0$, followed by L2 normalization. The perturbed centers are then combined to construct a shared batch-level reference direction:

$$\mathbf{c}_t = \mathrm{Norm}\left(\tilde{\boldsymbol{\mu}}^v + \tilde{\boldsymbol{\mu}}^t\right), \tag{20}$$

where $\mathbf{c}_t$ represents the instantaneous shared direction at training step t. Since mini-batch statistics may fluctuate during training, directly using $\mathbf{c}_t$ as the alignment target can introduce instability. Therefore, we maintain an exponential moving average to obtain a stabilized shared anchor direction:

$$\mathbf{a}_t = \mathrm{Norm}\left(m\mathbf{a}_{t-1} + (1-m)\mathbf{c}_t\right), \tag{21}$$

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where $m \in [0, 1)$ is a fixed EMA momentum coefficient and $\mathbf{a}_t$ denotes the shared anchor direction at training step t. The anchor is initialized as $\mathbf{a}_0 = \mathbf{c}_0$ and treated as a non-parametric target, with no gradient back-propagated through the EMA update. This provides a temporally smoothed reference direction while allowing the anchor to adapt to the evolving geometry of the shared embedding space during training. Unlike aligning one modality toward the other or toward a fixed prior direction, the shared anchor is jointly derived from both modality centers and does not designate either modality as the reference.

Based on the stabilized anchor direction, SADR constrains both modality centers toward the same reference direction:

$$\begin{aligned}\mathcal{L}_{\mathrm{SADR}} = w_v\big(1 - \cos(\boldsymbol{\mu}^v, \mathbf{a}_t)\big) \\ + w_t\big(1 - \cos(\boldsymbol{\mu}^t, \mathbf{a}_t)\big),\end{aligned} \tag{22}$$

where w_v and w_t are modality-specific weighting coefficients. By aligning the modality-level centers rather than individual samples, SADR reduces persistent directional discrepancy between image and text distributions. This preserves the local and instance-level discriminative structure learned by the contrastive objective.

E. Dual-Query Retrieval Inference

Although the training objectives improve both representation quality and embedding-space geometry, residual cross-modal discrepancy may still persist during inference. To address this, we propose a dual-query retrieval inference strategy that constructs two complementary SGPA queries from the original sentence embedding and a gap-aware text centroid, and adaptively fuses their corresponding similarity scores to produce a more robust final retrieval ranking.

Given an image i and a text query j, the first branch uses the global sentence embedding $\mathbf{f}_j^t$ as the SGPA query to aggregate patch-level visual tokens of image i, producing a query-conditioned visual representation $\tilde{\mathbf{f}}_{i|j}^v$. The corresponding similarity score is computed as $s_A(i,j) = \cos\left(\tilde{\mathbf{f}}_{i|j}^v, \mathbf{f}_j^t\right)$.

The global sentence embedding captures the overall semantics of the text query. However, it may not sufficiently emphasize the textual components that are most sensitive to residual cross-modal discrepancy. Therefore, the second branch constructs a gap-aware text centroid from word-level token embeddings and uses it as an alternative SGPA query. The residual modality gap direction is estimated as

$$\mathbf{g} = \mathrm{Norm}\left(\bar{\mathbf{f}}^v - \bar{\mathbf{f}}^t\right), \tag{23}$$

where $\bar{\mathbf{f}}^v$ and $\bar{\mathbf{f}}^t$ denote the mean visual and textual embeddings computed over the validation set. This direction is computed once after training and kept fixed during inference. For the j-th text query, let $\{\mathbf{f}_{j,l}^w\}_{l=1}^{L_j}$ denote its word-level token embeddings, with L_j being the number of valid tokens. The gap-aware token weight is computed as

$$\omega_l^j = \frac{\exp\left((\mathbf{f}_{j,l}^w \cdot \mathbf{g})/\tau_g\right)}{\sum_{l'=1}^{L_j} \exp\left((\mathbf{f}_{j,l'}^w \cdot \mathbf{g})/\tau_g\right)}, \tag{24}$$

where τ_g is a temperature parameter. The gap-aware text centroid is then obtained as

$$\mathbf{c}_j = \mathrm{Norm}\left(\sum_{l=1}^{L_j} \omega_l^j \mathbf{f}_{j,l}^w\right), \tag{25}$$

which is used as the second SGPA query to produce another query-conditioned visual representation $\tilde{\mathbf{f}}_{i|c_j}^v$, with the corresponding similarity score $s_B(i,j) = \cos\left(\tilde{\mathbf{f}}_{i|c_j}^v, \mathbf{f}_j^t\right)$.

Finally, the two complementary scores are adaptively fused to obtain the final retrieval score:

$$s(i,j) = \alpha s_A(i,j) + (1-\alpha) s_B(i,j), \tag{26}$$

where $\alpha \in [0, 1]$ is selected by grid search on the validation set and then fixed for final ranking. The first branch preserves global sentence-level semantics, the second branch emphasizes word-level textual components that are more sensitive to residual cross-modal discrepancy. Their fusion provides a more robust similarity estimate without relying on external linguistic tools. The dual-query inference strategy is applied consistently to both image-to-text and text-to-image retrieval by using the fused similarity matrix for final ranking.

IV. DATASETS AND EXPERIMENTAL SETTINGS

A. Datasets

We evaluate the proposed method on three publicly available remote sensing image-text retrieval benchmark datasets: RSICD [47], RSITMD [48], and UCM-Captions [49]. These datasets differ in scene type, annotation scale, and description style, providing a diverse evaluation environment for comprehensive assessment of the proposed method.

1) RSICD: One of the largest remote sensing image-text retrieval datasets, RSICD contains 10,921 images, each paired with five human-written descriptions, yielding 54,605 descriptions in total. The images cover 30 scene categories at a resolution of 224×224 pixels, including airports, harbors, farmlands, and residential areas. Its high semantic overlap and visually similar scenes make fine-grained cross-modal matching particularly challenging.

2) RSITMD: A fine-grained remote sensing image-text retrieval dataset containing 4,743 images and 23,715 descriptions across 32 scene categories. Compared with RSICD, RSITMD provides more detailed descriptions of local object attributes and spatial relationships, requiring stronger fine-grained semantic understanding.

3) UCM-Captions: Constructed from the UC Merced Land Use dataset, UCM-Captions contains 2,100 remote sensing images, each paired with five descriptions, yielding 10,500 texts. The images cover 21 land-use categories at a resolution of 256×256 pixels with a ground sampling distance of 0.3 meters, mainly depicting urban and suburban scenes.

B. Experimental Details

1) Data Preprocessing and Splitting: To ensure fair comparison with prior works, all datasets follow their respective standard splitting protocols. For each dataset, images are

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TABLE I
COMPARISON WITH SOTA METHODS ON THE RSITMD DATASET.

Methods	Backbone (image/text)	Params	Text Retrieval (%)			Image Retrieval (%)			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional									
LW-MCR-u [50]	ResNet-18, biGRU	1.65M	9.73	26.77	37.61	9.25	34.07	54.03	28.58
AMFMN-fusion [48]	ResNet-18, biGRU	35.94M	10.63	24.78	41.81	11.51	34.69	54.87	29.72
SWAN [10]	ResNet-50, biGRU	40M	13.35	32.15	46.90	11.24	40.40	60.60	34.11
MSITA [51]	ResNet-18, biGRU	48.14M	15.22	34.20	47.65	12.15	39.92	57.72	34.48
SCAT-PRG [52]	ResNet, Transformer	-	16.62	32.57	48.32	11.74	41.50	60.87	35.27
IEFT [53]	ResNet-50, Transformer	100.12M	15.48	37.61	51.40	11.10	37.99	58.75	35.38
KAMCL [8]	ResNet-18, biGRU	-	16.81	34.96	47.12	14.60	41.86	59.86	35.87
VGSGN [54]	ResNet-18, biGRU	54.31M	14.16	34.96	50.66	13.23	42.57	63.41	36.50
DOVE [19]	ResNet-50, biGRU	63M	16.81	36.80	49.93	12.20	49.93	66.50	37.73
MSA [55]	ResNet-50, biGRU	18.74M	15.93	38.50	50.88	14.96	45.22	61.86	37.89
PIR-ITR [22]	Swin-T, BERT	189M	18.36	42.04	55.53	13.36	44.47	61.73	39.25
CLIP-based									
Zero-shot CLIP [11]	CLIP(ViT-B/32)	0.00M	9.29	26.33	37.39	7.79	23.67	38.89	23.89
SkyCLIP [56]	CLIP(ViT-B/32)	0.00M	10.18	25.44	35.62	10.88	33.27	49.82	27.54
Adapter [57]	CLIP(ViT-B/32)	0.17M	13.75	27.64	39.96	12.89	40.09	59.91	32.37
CLIP-Adapter [32]	CLIP(ViT-B/32)	0.52M	12.83	28.84	39.05	13.30	40.20	60.06	32.38
Linear-probe CLIP [11]	CLIP(ViT-B/32)	0.26M	17.02	33.12	48.35	13.33	41.80	63.89	36.25
Cross-Modal Adapter [58]	CLIP(ViT-B/32)	0.16M	18.16	36.08	48.72	16.31	44.33	64.75	38.06
UniAdapter [59]	CLIP(ViT-B/32)	0.55M	19.86	36.32	51.28	17.54	44.89	56.46	39.23
SingleLanguage [60]	CLIP(ViT-B/32)	151M	19.69	40.26	54.42	17.61	49.73	66.59	41.38
PE-RSITR [14]	CLIP(ViT-B/32)	0.16M	23.67	44.07	60.36	20.10	50.63	67.97	44.47
Full-FT CLIP [11]	CLIP(ViT-B/32)	156M	24.16	47.12	61.28	20.40	50.53	68.54	45.33
HarMA [61]	CLIP(ViT-B/32)	5.93M	25.81	48.37	60.61	19.92	53.27	71.21	46.53
CLGSA [62]	CLIP(ViT-B/16)	-	22.52	52.39	**69.46**	**26.77**	48.23	60.39	46.63
Ours	**CLIP(ViT-B/32)**	**7.24M**	**29.65**	**53.54**	63.72	23.76	**56.77**	**74.03**	**50.24**

resized to 224×224 pixels and text descriptions are truncated to a maximum of 30 words. During training, we apply random resized cropping with a scale range of $[0.5, 1.0]$, random horizontal flipping, and RandAugment. During testing, only resizing and normalization are applied.

2) Implementation Details: All experiments are conducted in PyTorch on a single NVIDIA RTX 4090 GPU (24GB) using the LAION-2B pretrained CLIP ViT-B/32 backbone. The pretrained CLIP backbone remains frozen. For the baseline and all SGGA variants, the image/text adaptation layers, visual/text projection layers, and learnable logit scale are trainable; SGPA is additionally optimized when included, whereas BNDA and SADR introduce no additional trainable parameters. The adaptation layers and SGPA use a learning rate of 1×10^{-3}, while the projection layers and logit scale use 5×10^{-5}. We use AdamW with a weight decay of 0.01, cosine annealing, 8 training epochs, and a batch size of 256. Mini-batches are randomly sampled over image–caption pairs, and the contrastive loss uses diagonal targets without multi-positive masking; thus, captions of the same image that co-occur in a batch are treated as non-diagonal negatives under the same protocol for all variants.For BNDA, we set $k = 16$, $\tau = 0.08$, $\gamma = 0.75$, and $\epsilon = 10^{-8}$. All training hyperparameters, including λ_B and λ_S, are selected on the validation split. For dual-query inference, the modality-gap direction g is estimated once from the validation split after training, and the fusion coefficient α is selected by validation-set grid search.

C. Evaluation Metrics

We evaluate the proposed method using standard metrics for image-text retrieval in both Image-to-Text and Text-to-Image directions, including Recall@K (R@1, R@5, R@10) and mean recall (mR). Recall@K measures the proportion of queries for which the ground-truth match appears within the top-K retrieved results, formally defined as:

$$\text{R@K} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[\text{rank}(i) \le K] \tag{27}$$

where N is the total number of queries, $\text{rank}(i)$ denotes the rank of the ground-truth match for the i-th query in the retrieved results, and $\mathbf{1}[\cdot]$ is the indicator function. R@1 reflects the precise matching capability of the model, while R@5 and R@10 measure retrieval coverage under less stringent ranking criteria. mR is the overall average of R@1, R@5, and R@10 across both retrieval directions:

$$\text{mR} = \frac{1}{6}\sum_{K\in\{1,5,10\}}(\text{TR@}K + \text{IR@}K) \tag{28}$$

where TR@K and IR@K denote the Recall@K values in the Text-to-Image and Image-to-Text directions, respectively. mR is adopted as the primary criterion for model selection and performance comparison throughout this work.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Comparison With SOTA Methods

To validate the effectiveness and generalization capability of the proposed SGGA framework, we conduct comprehensive comparisons on three publicly available RSITR datasets against a range of baseline and SOTA methods. The compared methods include traditional cross-modal retrieval methods [50], [48], [10], [51], [52], [53], [8], [54], [19], [55], [22], and CLIP-based methods adapted for the remote sensing domain [11], [56], [57], [32], [58], [59], [60], [14], [61].

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TABLE II
COMPARISON WITH SOTA METHODS ON THE RSICD DATASET.

Methods	Backbone (image/text)	Params	Text Retrieval (%)			Image Retrieval (%)			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional									
LW-MCR-u [50]	ResNet-18, biGRU	1.65M	4.39	13.35	20.29	4.30	18.85	32.34	15.59
AMFMN-fusion [48]	ResNet-18, biGRU	35.94M	5.39	15.08	23.40	4.90	18.28	31.44	16.42
SWAN [10]	ResNet-50, biGRU	40M	7.41	20.13	30.86	5.56	22.26	37.41	20.61
MSITA [51]	ResNet-18, biGRU	48.14M	8.67	22.71	33.91	6.13	21.98	35.39	21.47
SCAT-PRG [52]	ResNet, Transformer	-	10.78	25.92	38.66	7.87	25.31	39.78	24.72
IEFT [53]	ResNet-50, Transformer	100.12M	8.11	27.45	43.25	6.26	24.00	39.98	24.84
KAMCL [8]	ResNet-18, biGRU	-	11.99	27.17	38.33	7.78	25.23	40.02	25.08
VGSGN [54]	ResNet-18, biGRU	54.31M	8.33	21.87	32.57	6.53	23.13	36.85	21.55
DOVE [19]	ResNet-50, biGRU	63M	8.66	22.35	34.95	6.04	23.95	40.35	22.72
MSA [55]	ResNet-50, biGRU	18.74M	9.52	25.25	35.32	6.55	24.43	39.63	23.45
PIR-ITR [22]	Swin-T, BERT	189M	10.89	26.17	37.79	7.17	25.07	41.06	24.69
CLIP-based									
Zero-shot CLIP [11]	CLIP(ViT-B/32)	0.00M	6.77	15.37	23.15	5.01	15.75	24.21	15.04
SkyCLIP [56]	CLIP(ViT-B/32)	0.00M	6.59	16.10	26.53	7.14	22.34	34.29	18.83
Adapter [57]	CLIP(ViT-B/32)	0.17M	8.73	24.73	37.81	8.43	26.02	43.33	24.84
CLIP-Adapter [32]	CLIP(ViT-B/32)	0.52M	7.11	19.48	31.01	7.67	24.87	39.73	21.65
Linear-probe CLIP [11]	CLIP(ViT-B/32)	0.26M	8.46	24.41	37.72	7.81	25.89	42.47	24.46
Cross-Modal Adapter [58]	CLIP(ViT-B/32)	0.16M	11.18	27.31	40.62	9.57	30.74	48.36	27.96
UniAdapter [59]	CLIP(ViT-B/32)	0.55M	12.65	30.81	42.74	9.61	30.06	47.16	28.84
SingleLanguage [60]	CLIP(ViT-B/32)	151M	10.70	29.64	41.53	9.14	28.96	44.59	27.42
PE-RSITR [14]	CLIP(ViT-B/32)	0.16M	14.13	31.51	44.78	11.63	33.92	50.73	31.12
Full-FT CLIP [11]	CLIP(ViT-B/32)	156M	13.54	30.83	43.46	11.55	33.14	49.83	30.39
HarMA [61]	CLIP(ViT-B/32)	5.93M	16.36	34.48	47.74	12.92	37.17	53.07	33.62
CLGSA [62]	CLIP(ViT-B/16)	-	11.75	34.07	50.12	14.38	36.73	45.35	32.07
Ours	**CLIP(ViT-B/32)**	**7.24M**	**17.57**	**37.51**	**50.59**	**13.89**	**37.77**	**53.52**	**35.14**

TABLE III
COMPARISON WITH SOTA METHODS ON THE UCM-CAPTIONS DATASET.

Methods	Backbone (image/text)	Params	Text Retrieval (%)			Image Retrieval (%)			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional									
LW-MCR-b [50]	ResNet-18, biGRU	1.65M	12.38	43.81	59.52	12.00	46.38	72.48	41.10
LW-MCR-d [50]	ResNet-18, biGRU	1.65M	15.24	51.90	62.86	11.90	50.95	75.24	44.68
LW-MCR-u [50]	ResNet-18, biGRU	1.65M	18.10	47.14	63.81	13.14	50.38	79.52	45.35
AMFMN-sim [48]	ResNet-18, biGRU	35.94M	14.76	49.52	68.10	13.43	51.81	76.48	45.68
AMFMN-soft [48]	ResNet-18, biGRU	35.94M	12.86	51.90	66.67	14.19	51.71	78.48	45.97
AMFMN-fusion [48]	ResNet-18, biGRU	35.94M	16.67	45.71	68.57	12.86	53.24	79.43	46.08
MSITA [51]	ResNet-18, biGRU	48.14M	16.86	49.33	73.33	14.29	57.16	91.58	50.43
MSA [55]	ResNet-50, biGRU	18.74M	13.33	59.05	77.14	13.14	57.52	92.48	52.11
CLIP-based									
Zero-shot CLIP [11]	CLIP(ViT-B/32)	0.00M	10.95	34.76	55.71	7.33	35.14	54.57	33.08
CLIP-Adapter [32]	CLIP(ViT-B/32)	0.52M	9.83	42.52	66.79	14.61	51.03	84.14	44.82
Linear-probe CLIP [11]	CLIP(ViT-B/32)	0.26M	13.33	52.71	77.62	14.43	59.42	90.28	51.30
Full-FT CLIP [11]	CLIP(ViT-B/32)	156M	17.14	55.24	79.52	13.90	56.95	91.81	52.43
Adapter [57]	CLIP(ViT-B/32)	0.17M	10.17	49.73	80.08	18.20	62.47	95.77	52.74
SingleLanguage [60]	CLIP(ViT-B/32)	151M	19.04	53.33	77.61	19.33	64.00	91.42	54.12
Cross-modal Adapter [58]	CLIP(ViT-B/32)	0.16M	13.77	50.57	81.41	17.43	61.45	95.48	53.62
UniAdapter [59]	CLIP(ViT-B/32)	0.55M	14.46	55.19	83.95	16.74	61.43	95.76	54.59
PE-RSITR [14]	CLIP(ViT-B/32)	0.16M	22.71	55.81	80.33	18.82	62.84	93.72	55.71
Ours	**CLIP(ViT-B/32)**	**7.24M**	**27.14**	**64.29**	**83.81**	**19.05**	**66.19**	**96.67**	**59.52**

1) Results on RSITMD dataset: Table I presents the comparison results on the RSITMD dataset, which emphasizes fine-grained object attributes and spatial relationships and therefore requires strong local semantic alignment. SGGA achieves the best overall mR of 50.24%, outperforming HarMA by 3.71% under the same ViT-B/32 backbone, with R@1 improvements of 3.84 percentage points in both retrieval directions. Although CLGSA obtains higher Text Retrieval R@10 and Image Retrieval R@1 scores, it uses the stronger ViT-B/16 backbone, which produces approximately four times as many visual tokens as ViT-B/32. Nevertheless, SGGA surpasses CLGSA by 3.61% in overall mR, demonstrating the effectiveness of the proposed framework.

2) Results on RSICD dataset: Table II presents the comparison results on the RSICD dataset. RSICD covers 30 scene categories with substantial visual similarity and semantic overlap across images, presenting significant challenges for cross-modal discriminability. SGGA achieves the best overall mR of 35.14% among all compared methods, outperforming HarMA by 1.52% under the same ViT-B/32 backbone, with R@1 improvements of 1.21% and 0.97% in the Image-to-Text and Text-to-Image directions, respectively. CLGSA achieves a slightly higher Image Retrieval R@1 (14.38% vs. 13.89%) with its stronger ViT-B/16 backbone, but SGGA surpasses it in overall mR by 3.07% (35.14% vs. 32.07%), indicating stronger holistic retrieval capability under high semantic overlap.

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TABLE IV
ABLATION STUDY OF KEY COMPONENTS IN SGGA.

Method	SGGA Components			Text Retrieval			Image Retrieval			mR
	SGPA	BNDA	SADR	R@1	R@5	R@10	R@1	R@5	R@10	
Baseline+	×	×	×	25.88	48.23	62.39	21.02	55.22	72.35	47.51
	✓	×	×	27.88	51.55	63.72	23.85	56.06	73.76	49.47
	×	✓	×	27.88	52.43	64.60	23.50	56.15	73.23	49.63
	×	×	✓	28.32	52.43	63.94	23.19	56.19	72.96	49.51
	×	✓	✓	28.10	52.43	64.60	23.45	56.19	73.23	49.67
	✓	×	✓	28.32	52.88	63.72	23.01	56.46	74.12	49.75
	✓	✓	×	28.54	52.88	**64.60**	23.32	56.59	**74.16**	50.01
Ours(SGGA)	✓	✓	✓	**29.65**	**53.54**	63.72	**23.76**	**56.77**	74.03	**50.24**

TABLE V
SINGLE-SOURCE ZERO-SHOT CROSS-DATASET TRANSFER RESULTS.

Training	Unseen	Method	Text Retrieval (%)			Image Retrieval (%)			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
RSITMD	RSICD	CLIP	14.09	29.73	42.73	11.22	33.50	49.04	30.05
		Ours	**33.21**	**53.61**	**65.23**	**25.32**	**51.64**	**65.67**	**49.12**
	UCM	CLIP	14.76	43.81	**65.24**	**14.67**	44.86	70.76	42.35
		Ours	**16.90**	**43.86**	**65.24**	13.52	**48.67**	**77.24**	**44.24**
RSICD	RSITMD	CLIP	21.46	45.13	59.73	18.76	49.16	67.65	43.65
		Ours	**32.74**	**53.10**	**67.48**	**23.72**	**58.41**	**74.78**	**51.70**
	UCM	CLIP	14.76	**45.71**	65.24	11.90	**49.05**	**77.71**	44.06
		Ours	**18.38**	45.24	**69.05**	**13.52**	**49.05**	76.10	**45.22**
UCM	RSITMD	CLIP	**11.06**	**25.22**	37.61	10.31	31.73	44.20	26.69
		Ours	5.97	21.90	**38.72**	**11.50**	**36.15**	**51.24**	**27.58**
	RSICD	CLIP	6.68	15.83	24.52	5.53	17.58	28.42	16.43
		Ours	**7.32**	**17.38**	**26.44**	**6.08**	**19.93**	**32.61**	**18.29**

3) Results on UCM-Captions dataset: Table III presents the comparison results on the UCM-Captions dataset, which contains 21 land-use categories with a relatively small scale and well-defined category boundaries. SGGA achieves SOTA performance, with R@1, R@5, and R@10 of 27.14%, 64.29%, and 83.81% for Image-to-Text retrieval, and 19.05%, 66.19%, and 96.67% for Text-to-Image retrieval, yielding an overall mR of 59.52%. Compared with the previous best-performing method, SGGA improves mR by 3.81%, including a 4.43% gain in Image-to-Text R@1 and consistent improvements across both retrieval directions, demonstrating its effectiveness across datasets with different scales and category structures.

B. Generalization Evaluation

1) Cross-Dataset Transfer Evaluation: To further evaluate cross-dataset generalization, we conduct single-source zero-shot experiments by training on one source dataset and directly evaluating on the target test set. As shown in Table V, our method outperforms the source-trained CLIP baseline in all six transfer directions, increasing average mR from 33.87% to 39.36%, a gain of 5.49 points. The largest gains occur for RSITMD-to-RSICD and RSICD-to-RSITMD transfer, with improvements of 19.07 and 8.05 points, respectively, likely due to their closer visual-semantic distributions and greater overlap in aerial-scene concepts. Although gains involving UCM-Captions are smaller (0.89–1.89 points), our method consistently improves performance under larger distribution shifts. These results demonstrate improved cross-dataset generalization across both related and more distinct domains.

TABLE VI
GENERALIZATION OF SGGA TO REMOTECLIP VIT-B/32 ON THREE RSITR DATASETS.

Dataset	Method	I2T R@1	I2T R@5	I2T R@10	T2I R@1	T2I R@5	T2I R@10	mR
RSITMD	RemoteCLIP	27.88	50.66	65.71	22.17	56.46	73.41	49.38
	+ SGGA	**29.87**	**52.57**	**65.95**	**24.92**	**57.93**	**74.42**	**50.78**
RSICD	RemoteCLIP	17.02	37.97	51.51	13.71	37.11	54.25	35.26
	+ SGGA	**18.57**	**38.06**	**52.15**	**14.00**	**39.49**	**55.86**	**36.36**
UCM	RemoteCLIP	**20.48**	59.85	83.33	18.67	61.52	94.29	56.36
	+ SGGA	19.52	**61.90**	**83.81**	**18.76**	**65.52**	**96.19**	**57.62**

TABLE VII
QUERY-RELEVANCE ANALYSIS OF CLIP VISUAL REPRESENTATIONS.

Mask	Patch Perturbation (ΔmR)			CLS Attention Mass (%)	
	Low-rel.	Random	High-rel.	Low-rel.	Uniform
10%	+1.14	$+0.24 \pm 0.28$	-1.35	15.64	10.00
20%	+1.91	$+0.56 \pm 0.26$	-3.20	27.43	20.00
30%	+2.38	$+0.57 \pm 0.32$	-4.88	37.95	30.00

2) Domain-Specific Pretrained Backbone: The main results use SGGA with the general-domain CLIP ViT-B/32 backbone. To verify its effectiveness under domain-specific initialization, we further evaluate it on RemoteCLIP ViT-B/32 [12]. As shown in Table VI, SGGA consistently improves mR over the RemoteCLIP baseline across all three datasets, with gains of 1.40, 1.10, and 1.26 points on RSITMD, RSICD, and UCM-Captions, respectively. Most recall metrics also improve, except I2T R@1 on UCM-Captions, where the small dataset size and high category redundancy may cause greater top-1 variance; R@5, R@10, and mR remain consistently improved. These results show that SGGA is complementary to domain-specific vision–language pretraining.

3) Analysis of Query-Dependent Visual Relevance: To examine whether query-independent visual representations retain local content weakly related to a text query, we conduct patch perturbation and CLS attention analyses on the trained CLIP baseline on RSITMD. Patch relevance is ranked by cosine similarity to the corresponding target text; therefore, this target-text-guided masking is an oracle-based diagnostic used only for analysis. As shown in Table VII, masking the least relevant 10%, 20%, and 30% of patches improves the diagnostic mR by 1.14, 1.91, and 2.38 points, respectively, while masking the most relevant patches decreases it by 1.35–4.88 points. Random masking produces much smaller changes,

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TABLE VIII
ABLATION STUDY OF SGPA MODULE COMPONENTS

Model Variant	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
w/o SGPA	28.1	52.43	64.6	23.45	56.19	73.23	49.67
w/o Tanh Gate	28.76	53.10	63.72	23.01	56.42	74.03	49.84
w/o Bottleneck MLP	29.20	53.98	63.50	23.76	56.42	74.03	50.15
Ours	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**

TABLE IX
ABLATION STUDY OF BNDA COMPONENTS

Model Variant	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
w/o BNDA	28.32	52.88	63.72	23.01	56.46	74.12	49.75
BNDA ($\gamma = 0$)	28.98	53.28	63.59	23.45	56.68	73.91	49.98
+ Text to Image	29.20	52.21	**65.27**	**23.81**	56.33	73.45	50.04
+ Image to Text	28.98	**53.76**	63.94	23.63	56.55	73.85	50.12
Ours (Bidirectional)	**29.65**	53.54	63.72	23.76	**56.77**	74.03	**50.24**

TABLE X
ABLATION STUDY OF DUAL-QUERY CONSTRUCTION

Model Variant	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
Original query only	29.12	50.27	62.65	23.70	56.63	73.56	49.32
Uniform token pooling	28.32	49.12	62.61	22.70	**56.95**	72.26	48.66
Learned attention pooling	28.99	49.67	**63.94**	23.14	56.46	73.27	49.25
Dual-query w/o gap correction	29.10	51.78	62.73	23.02	56.73	73.17	49.42
Dual-query ($-g$)	28.36	52.12	62.95	22.74	56.59	72.35	49.18
Dual-query (Random g)	29.12 ± 0.10	52.63 ± 0.21	62.41 ± 0.24	22.86 ± 0.08	56.72 ± 0.06	72.36 ± 0.05	49.35 ± 0.068
Ours (Dual-query $+g$)	**29.65**	**53.54**	63.72	**23.76**	56.77	**74.03**	**50.24**

TABLE XI
ABLATION STUDY OF ANCHOR CONSTRUCTION AND STABILIZATION IN SADR.

Model Variant	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
w/o SADR	28.54	52.88	64.6	23.32	56.59	74.16	50.01
Fixed Batch-Mean Anchor	28.76	53.32	64.16	23.5	56.19	73.98	49.99
Low-Momentum	28.98	53.76	63.72	23.67	56.73	74.12	50.16
High-Momentum	29.2	53.54	63.72	23.76	56.77	74.03	50.17
Ours (Gaussian + EMA)	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**

suggesting an association with patch relevance. Moreover, the least relevant patches consistently receive more CLS attention mass than their uniform-reference levels. These results suggest that query-independent CLS aggregation retains non-negligible responses from query-low-relevance regions, which may interfere with query-specific matching and thus support the motivation for SGPA.

C. Ablation Studies

To comprehensively evaluate the effectiveness and contribution of each component, we conduct a series of ablation experiments on the RSITMD dataset.

1) Ablation Study of SGGA Key Components: To evaluate the contribution of each core module in SGGA, we use CLIP as the baseline and analyze the individual and combined effects of SGPA, BNDA, and SADR in Table IV. All variants outperform the baseline (mR: 47.51%), confirming the effectiveness of the proposed modules. Individually, SGPA, BNDA, and SADR improve mR to 49.47%, 49.63%, and 49.51%, respectively, by enhancing query-conditioned visual representation, cross-modal neighborhood consistency, and shared-space alignment. Among two-module combinations, SGPA+BNDA achieves the highest mR of 50.01%, followed by SGPA+SADR at 49.75% and BNDA+SADR at 49.67%, indicating strong complementarity between SGPA and BNDA. The full model achieves the best mR of 50.24%, demonstrating that the three modules provide complementary benefits for cross-modal matching.

2) Ablation Study of SGPA Module Components: To analyze the contribution of each SGPA design component, we conduct targeted ablations in Table VIII. Compared with the w/o SGPA baseline (mR: 49.67%), all variants achieve higher performance, confirming the effectiveness of SGPA in constructing query-relevant visual representations. Removing the $\tanh$ gate reduces mR to 49.84%, showing that the gated residual update helps stably incorporate text-guided information while preserving pretrained visual features. Removing the bottleneck MLP projection lowers mR to 50.15%, indicating that the nonlinear projection facilitates more effective query–key interactions in cross-attention.

3) Ablation Study of Directionality in BNDA: To validate the effectiveness of bidirectional neighborhood consistency in BNDA, we compare variants in Table IX, where the baseline is the full model without BNDA (SGPA+SADR, mR: 49.75%). Using bidirectional neighborhood alignment without semantic rescaling ($\gamma = 0$) improves mR to 49.98%, showing that distribution alignment is effective on its own. The unidirectional Text-to-Image and Image-to-Text constraints improve mR to 50.04% and 50.12%, respectively, but both remain below the full bidirectional constraint, which achieves the best mR of 50.24%. These results confirm the complementary benefits of semantic rescaling and bidirectional neighborhood alignment.

4) Ablation Study of Dual-Query Construction: To analyze the contribution of the gap direction in dual-query retrieval, we compare alternative pooling and directional variants in Table X. Uniform token pooling and learned attention pooling obtain mR values of 48.66% and 49.25%, respectively, while dual-query retrieval without gap correction achieves 49.42%. Reversing the gap direction ($-g$) or replacing it with a random direction yields 49.18% and 49.35 ± 0.068%, respectively. In contrast, the proposed $+g$ direction achieves the best mR of 50.24%, indicating that the gain is associated with the specifically defined gap direction.

5) Ablation Study of Anchor Construction and Stabilization in SADR: To analyze the contributions of anchor construction and temporal stabilization in SADR, we compare several anchor variants in Table XI. Directly using the current batch-mean direction as the anchor slightly decreases mR from 50.01% to 49.99%. Applying EMA stabilization without Gaussian perturbation improves performance to 50.16% and 50.17% under low- and high-momentum settings, respectively. Our complete SADR further introduces Gaussian perturbation before fixed-momentum EMA stabilization and achieves the best mR of 50.24%. These results indicate that temporal smoothing improves the stability of the shared anchor, while Gaussian perturbation provides an additional benefit beyond EMA stabilization alone.

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TABLE XII
COMPARISON OF SGPA WITH ALTERNATIVE AGGREGATION DESIGNS

Method	Head / Total Trainable Params	MACs / Pair	Latency (μs/Pair)	Text Retrieval (%)			Image Retrieval (%)			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Mean Pooling	1.0496M / 7.2407M	0.0005M	0.029	27.43	49.34	60.40	23.01	55.66	72.92	48.13
Cross-Attention	1.0506M / 7.2417M	0.314M	0.548	27.21	48.89	**63.94**	23.72	55.53	72.79	48.68
SGPA	1.0506M / 7.2417M	0.314M	0.586	**27.88**	**51.55**	63.72	**23.85**	**56.06**	**73.76**	**49.47**

TABLE XIII
EFFECT OF NEIGHBORHOOD SCALE IN BNDA

topk	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
8	28.98	53.76	63.5	23.36	56.59	74.16	50.06
16	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**
24	28.76	53.32	63.72	23.54	56.73	74.07	50.02
32	28.98	53.32	63.72	23.72	56.81	74.03	50.10

TABLE XIV
SENSITIVITY ANALYSIS OF γ IN BNDA.

γ	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0	28.98	53.28	63.59	23.45	56.68	73.91	49.98
0.25	28.95	53.33	63.68	23.85	56.73	**74.07**	50.11
0.50	29.27	53.49	**63.72**	**23.94**	**56.81**	74.05	50.20
0.75	**29.65**	**53.54**	**63.72**	23.76	56.77	74.03	**50.24**
1.00	29.36	53.32	63.70	23.85	56.73	74.01	50.16

TABLE XV
EFFECT OF ASYMMETRIC MODALITY WEIGHTING IN SADR

w_v	w_t	Text Retrieval (%)			Image Retrieval (%)			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
0.0	1.0	29.2	53.54	63.72	23.67	56.77	73.08	50.15
1.0	**1.0**	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**
1.5	1.0	29.42	53.54	63.72	23.76	56.77	74.03	50.21
1.0	0.0	29.42	53.54	63.72	23.72	56.73	74.16	50.21
1.0	1.5	29.42	53.54	63.72	23.76	56.81	73.98	50.21

TABLE XVI
SENSITIVITY TO GAUSSIAN NOISE SCALE IN SADR

σ	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0	29.29	53.15	62.61	22.83	56.80	73.46	49.86
0.5	29.32	53.12	63.05	23.01	56.53	73.95	50.00
1.0	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**
1.5	29.32	52.89	63.05	22.88	56.31	73.95	49.90
2.0	29.42	52.97	63.05	22.92	56.31	73.95	49.93

6) Comparison with Alternative Aggregation Designs: To isolate the design-specific contribution of SGPA, we compare it with a simple Mean Pooling baseline and a parameter-matched text-conditioned Cross-Attention reranker under the same training and pairwise inference protocol, as shown in Table XII. The Cross-Attention baseline is configured to match SGPA in trainable parameter count and pairwise computation. Mean Pooling and Cross-Attention achieve mR values of 48.13% and 48.68%, respectively. SGPA further improves mR to 49.47%, outperforming the parameter-matched Cross-Attention baseline by 0.79 points, supporting the design-specific effectiveness of SGPA under comparable model capacity and computation.

D. Parameter sensitivity analysis

1) Effect of Neighborhood Scale in BNDA: To analyze the sensitivity of BNDA to neighborhood scale, we evaluate model performance across $k \in \{8, 16, 24, 32\}$, with results reported in Table XIII. Performance degrades at both extremes: a small $k = 8$ limits topological coverage and reduces the ability to preserve consistent neighborhood relationships across modalities, while larger values of $k = 24$ and $k = 32$ introduce semantically irrelevant samples that add noise to the neighborhood consistency constraint. $k = 16$ achieves the best mR of 50.24%, indicating that this scale provides sufficient coverage of semantically related samples while avoiding interference from distant neighbors.

2) Sensitivity Analysis of γ in BNDA: To evaluate the effect of semantic rescaling strength, we vary γ from 0 to 1.0 in Table XIV. Setting $\gamma = 0$ disables semantic rescaling and yields an mR of 49.98%. Performance improves as γ increases and reaches the best mR of 50.24% at $\gamma = 0.75$, while a larger value of 1.0 slightly decreases mR to 50.16%. These results indicate that moderate semantic rescaling is beneficial and that BNDA is relatively stable across different γ values.

3) Effect of Asymmetric Modality Weighting in SADR: To analyze the influence of modality-specific weighting in SADR, we conduct sensitivity experiments on (w_v, w_t) with results reported in Table XV. The symmetric setting $(w_v = 1.0, w_t = 1.0)$ achieves the best mR of 50.24%. Disabling the visual-side constraint $(w_v = 0.0)$ decreases mR to 50.15%, while disabling the text-side constraint $(w_t = 0.0)$ decreases mR to 50.21%, confirming that anchor-direction regularization on both modalities is necessary for balanced alignment. Increasing either weight beyond 1.0 also leads to marginal performance degradation, suggesting that maintaining symmetric regularization strength is the most stable configuration for shared-anchor alignment.

4) Effect of Gaussian Noise Scale in SADR: To analyze the influence of Gaussian perturbation in SADR, we evaluate $\sigma \in \{0, 0.5, 1.0, 1.5, 2.0\}$, with results reported in Table XVI. The setting $\sigma = 1.0$ achieves the best mR of 50.24%, while both weaker and stronger perturbations lead to slight performance degradation. This indicates that moderate Gaussian noise can

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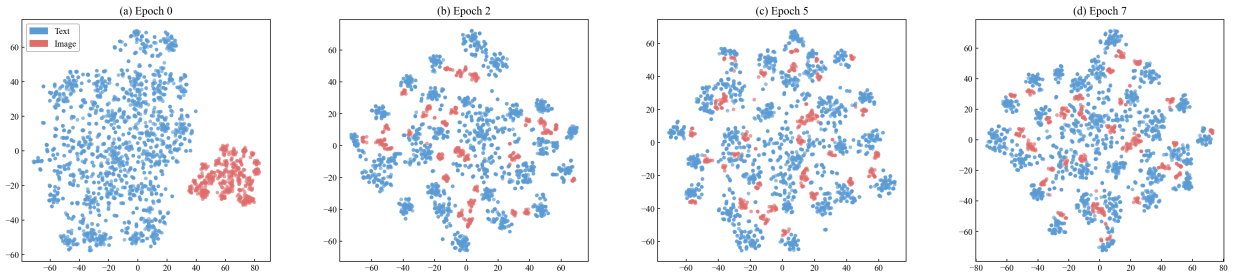


Fig. 6. t-SNE visualization of the cross-modal embedding space at Epoch 0, 2, 5, and 7. Blue and red points represent text and image embeddings, respectively, illustrating the progressive alignment of the two modalities toward a unified semantic space throughout training.

TABLE XVII
SENSITIVITY TO EMA MOMENTUM IN SADR

m	Text Retrieval (%)			Image Retrieval (%)			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0.95	29.88	52.67	63.05	22.96	56.58	73.81	49.83
0.99	29.32	55.39	63.55	22.96	56.27	73.95	49.91
0.995	29.34	55.42	63.55	22.93	56.40	73.95	50.02
0.999	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**

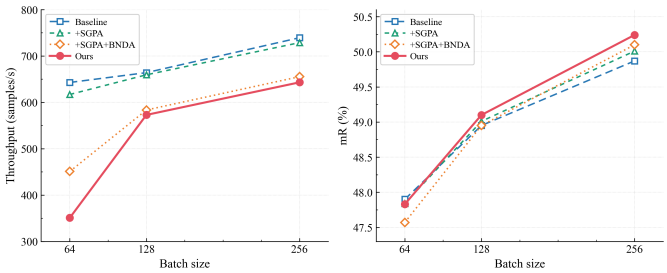


Fig. 7. Comparison of training throughput (left) and retrieval mR (right) across four model variants under varying batch sizes. Ours denotes the full SGGA framework.

improve the robustness of the shared anchor against mini-batch fluctuations, whereas excessive perturbation may disturb the underlying modality-center directions.

5) Effect of EMA Momentum in SADR: To evaluate the sensitivity of SADR to the EMA momentum coefficient, we vary $m \in \{0.95, 0.99, 0.995, 0.999\}$ while keeping all other settings unchanged, with results reported in Table XVII. The corresponding mR values are 49.83%, 49.91%, 50.02%, and 50.24%, respectively. The performance remains relatively stable across the evaluated settings, with $m = 0.999$ achieving the best result. We therefore use $m = 0.999$ in all subsequent experiments.

6) Scalability Analysis across Batch Sizes: To analyze training efficiency and retrieval performance under different batch sizes, we compare model variants on RSITMD, as shown in Fig. 7. Throughput increases consistently as the batch size grows from 64 to 256 for all variants due to improved batch-parallel efficiency. The full SGGA framework has slightly lower throughput because of the additional computations introduced by SGPA and BNDA, but follows a similar scaling trend, indicating good batch-level scalability. In terms of retrieval performance, SGGA achieves the best mR across all batch sizes, reaching 50.24% at batch size 256 while retaining a clear advantage at batch size 64, showing that its gains do not depend on large-batch training and remain effective under limited computational resources.

E. Qualitative Analysis

1) Embedding Space Visualization: To further analyze the geometric evolution of the shared embedding space during training, we visualize image and text embeddings from the RSITMD test set using t-SNE at Epochs 0, 2, 5, and 7, as shown in Fig. 6. At Epoch 0, the two modalities form clearly separated clusters, indicating substantial cross-modal distributional inconsistency. As training progresses, the image embeddings gradually approach the text distribution, with partial interleaving at Epoch 5 and substantial overlap by Epoch 7. This evolution shows that the collaborative optimization of SGPA, BNDA, and SADR progressively reduces directional discrepancy and neighborhood inconsistency between modalities, providing geometric evidence for the effectiveness of the proposed alignment objectives.

2) Quantitative Geometry Analysis: For each image I_i with five captions $\{T_{i,j}\}_{j=1}^{5}$, we average the corresponding caption-conditioned visual and textual representations:

$$\bar{\mathbf{v}}_i = \mathrm{Norm}\left(\frac{1}{5}\sum_{j=1}^{5}\mathrm{SGPA}(I_i, T_{i,j})\right), \tag{29}$$

$$\bar{\mathbf{t}}_i = \mathrm{Norm}\left(\frac{1}{5}\sum_{j=1}^{5}\mathbf{t}_{i,j}\right). \tag{30}$$

This yields $N = 452$ paired image–text representations on the RSITMD test split; the baseline uses its query-independent global visual representation. Center Dist. is the Euclidean distance between modality means. Jaccard@20 measures the overlap of corresponding cosine top-20 intra-modal neighborhoods excluding self-neighbors, and Spearman is the rank correlation between the upper-triangular entries of $\bar{\mathbf{V}}\bar{\mathbf{V}}^\top$ and $\bar{\mathbf{T}}\bar{\mathbf{T}}^\top$. Alignment MSE is $\frac{1}{N}\sum_{i=1}^{N}\|\bar{\mathbf{v}}_i - \bar{\mathbf{t}}_i\|_2^2$. Uniformity is $U(X) = \log \mathbb{E}_{x\neq y}\exp(-2\|x-y\|_2^2)$, computed separately for $\bar{\mathbf{V}}$ and $\bar{\mathbf{T}}$ and then averaged. Isotropy is the similarly averaged normalized spectral entropy of the centered covariance eigenvalues. Hubness is the Gini coefficient of target occurrence

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TABLE XVIII
QUANTITATIVE ANALYSIS OF CROSS-MODAL EMBEDDING GEOMETRY ON RSITMD.

Method	SGGA Components			Center Dist.↓	Jaccard@20 (%)↑	Spearman↑	Alignment MSE↓	Uniformity↓	Isotropy↑	T2I / I2T Hubness↓
	SGPA	BNDA	SADR							
Baseline+	×	×	×	0.5133 ± 0.0037	29.159 ± 0.163	0.3265 ± 0.0040	1.2932 ± 0.0034	-3.5981 ± 0.0058	0.7013 ± 0.0011	0.1915 ± 0.0083 / 0.1891 ± 0.0024
	✓	×	×	0.2982 ± 0.0108	30.314 ± 0.533	0.3142 ± 0.0147	1.2252 ± 0.0041	-3.6051 ± 0.0054	0.7122 ± 0.0013	0.1814 ± 0.0073 / 0.1822 ± 0.0024
	✓	×	✓	0.2844 ± 0.0051	31.210 ± 0.660	0.3320 ± 0.0148	1.1824 ± 0.0027	-3.7427 ± 0.0014	**0.7130 ± 0.0013**	0.1713 ± 0.0048 / 0.1753 ± 0.0022
	✓	✓	×	0.2947 ± 0.0086	31.354 ± 0.357	0.3366 ± 0.0133	1.1806 ± 0.0031	-3.7639 ± 0.0048	0.7127 ± 0.0010	0.1709 ± 0.0075 / 0.1737 ± 0.0049
Ours (SGGA)	✓	✓	✓	**0.2798 ± 0.0081**	**32.386 ± 0.447**	**0.3457 ± 0.0137**	**1.1697 ± 0.0019**	**-3.7654 ± 0.0053**	**0.7130 ± 0.0010**	**0.1622 ± 0.0028 / 0.1672 ± 0.0043**

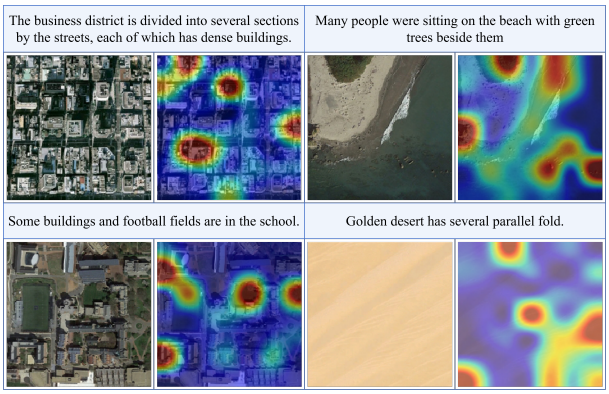


Fig. 8. Visualization of query-conditioned attention responses generated by SGPA on representative RSITMD samples. Each example shows the text query, original image, and corresponding attention map, where warmer regions indicate stronger responses. Different queries induce distinct spatial attention patterns over semantically relevant visual regions.

counts in cosine top-10 retrieval, reported separately for T2I and I2T. Fig. 6 qualitatively illustrates the progressive cross-modal alignment during training, which is further quantified by the geometry metrics in Table XVIII. SGPA alone reduces the cross-modal center distance from 0.5133 to 0.2982, while the full model further decreases it to 0.2798, indicating reduced directional discrepancy between modalities. Adding BNDA improves Jaccard@20 from 30.314% to 31.354% and Spearman correlation from 0.3142 to 0.3366, indicating more consistent image- and text-defined neighborhoods. The full model achieves the best overall geometry, with lower alignment error and hubness, stronger neighborhood consistency, and high isotropy. Removing either BNDA or SADR degrades the corresponding metrics, supporting their complementary contributions to cross-modal geometry.

3) Query-Conditioned Attention Visualization: To further examine how SGPA establishes intermediate cross-modal semantic correspondence, we visualize its text-conditioned attention responses on representative RSITMD samples in Fig. 8. Given each text query, the multi-head attention weights in SGPA are aggregated and mapped back to the spatial patch grid. The resulting maps show that SGPA produces query-dependent responses over semantically relevant image regions. For example, queries describing dense buildings and football fields emphasize the corresponding built-up structures and field regions, while descriptions involving beaches, trees, and desert patterns induce different spatial responses. These observations qualitatively demonstrate that textual semantics directly modulate visual feature aggregation before the final similarity computation, providing intermediate-level evidence



Fig. 9. Retrieval result visualization on the RSITMD dataset. (Top) Image-to-Text retrieval: Top-5 retrieved captions per query image, with correct and incorrect matches shown in green and red text respectively. (Bottom) Text-to-Image retrieval: Top-5 retrieved images per query caption, with correct and incorrect matches indicated by green and red borders respectively.

for the query-conditioned representation mechanism of SGPA.

4) Bidirectional Retrieval Results: We evaluate the bidirectional retrieval performance of SGGA on RSITMD, as shown in Fig. 9. For image-to-text retrieval, SGGA accurately captures key semantics and retrieves relevant descriptions for distinctive targets such as water storage tanks, railway stations, and blue-roofed factories. For text-to-image retrieval, it effectively distinguishes visually similar scenes using land-cover and spatial cues. Even when strict matches are not retrieved, the returned results remain semantically consistent with the queries, demonstrating the robustness of SGGA in complex remote sensing scenarios.

F. Efficiency and Scalability Analysis

1) Training and Model Efficiency: To evaluate computational overhead, we analyze component-wise training efficiency and model complexity under a unified profiling protocol. The variants in Table XIX(a) correspond to those reported in Table VII. As shown in Table XIX(a), progressively adding SGPA, BNDA, and SADR increases training time from 3.10 to 3.56 minutes and reduces throughput from 739.4 to 643.2 samples/s, while peak GPU memory changes by only 0.10 GB.

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TABLE XIX
TRAINING AND MODEL EFFICIENCY ON RSITMD.

(a) Training efficiency of different model variants.

Model Variant	Training Time (min)	Peak GPU Memory (GB)	Throughput (samples/s)	mR (%)
Baseline	3.10	13.96	739.4	47.51
+SGPA	3.14	14.06	728.8	49.47
+SGPA+BNDA	3.49	13.99	655.4	50.01
Ours (SGGA)	3.56	14.06	643.2	**50.24**

(b) Efficiency comparison with representative methods.

Method	Total / Trainable Params (M)	FLOPs (G/pair)	Image Encoding Latency (ms/sample)	Text Encoding Latency (ms/sample)	mR (%)
Full-FT CLIP	156.81 / 156.81	15.28	6.718 ± 0.057	7.041 ± 0.246	45.33
PE-RSITR	156.97 / 0.16	15.52	7.960 ± 0.045	8.285 ± 0.043	44.47
HarMA	157.21 / 5.93	15.60	10.643 ± 0.267	14.672 ± 0.286	46.53
Ours (SGGA)	157.86 / 7.24	15.36	6.988 ± 0.063	7.348 ± 0.636	**50.24**

TABLE XX
ACCURACY AND SCALABILITY OF TWO-STAGE SGPA ON RSITMD.

(a) Retrieval performance under different candidate sizes.

Method	K	Text Retrieval (%)			Image Retrieval (%)			mR (%)
		R@1	R@5	R@10	R@1	R@5	R@10	
Global-only Retrieval	–	26.11	51.55	62.17	23.98	55.93	73.63	48.89
Two-stage SGPA	10	29.65	53.32	63.72	23.76	56.77	73.72	50.15
Two-stage SGPA	**20**	**29.65**	**53.54**	**63.72**	**23.76**	**56.77**	**74.03**	**50.24**
Exhaustive SGPA	ALL	29.65	53.54	63.72	23.76	56.77	74.03	50.24

(b) Online latency and memory scalability.

Gallery Size	Method	Latency (ms/query)	Feature Cache (MB)	Online Memory Increment (MB)
452	Exhaustive SGPA	0.259 ± 0.002	89.2	46.1
	Two-stage SGPA	0.309 ± 0.002	**4.8**	**6.1**
5K	Exhaustive SGPA	1.692 ± 0.001	986.3	509.5
	Two-stage SGPA	0.311 ± 0.002	**13.7**	**6.1**
20K	Exhaustive SGPA	6.619 ± 0.007	3945.3	2040.6
	Two-stage SGPA	0.313 ± 0.001	**43.0**	**20.0**

(c) Actual HNSW retrieval scalability.

Gallery Size	HNSW Search (ms/query)	Recall@20 vs. Exact (%)	Index Memory (MB)
5K	0.082	100.00	11.08
20K	0.092	100.00	44.33

Compared with the baseline, the full model increases training time by 14.8% while improving mR by 2.73 percentage points. Table XIX(b) compares model complexity and encoding efficiency with representative methods. Our method uses only 7.24M trainable parameters, 95.4% fewer than Full-FT CLIP, while increasing FLOPs by only 0.08G per image-text pair. Its image and text encoding latencies remain close to Full-FT CLIP and lower than PE-RSITR and HarMA, indicating limited computational overhead.

2) Two-Stage Inference and Scalability: Exhaustive SGPA applies query-conditioned visual modeling to all N gallery images, with a complexity of $O(NN_pD)$. We instead retrieve the top-K candidates using cached global embeddings and rerank them with SGPA, reducing online complexity to $O(ND) + O(KN_pD)$ while remaining compatible with standard ANN indices. The candidate size K is selected on the validation split, where $K = \{5, 10, 20, 50, 100\}$ yields mR values of 33.15%, 35.28%, 35.46%, 35.30%, and 35.28%, respectively. We therefore fix $K = 20$ for all test and scalability evaluations. As shown in Table XX(a), $K = 20$ matches exhaustive SGPA with an mR of 50.24%. We construct 5K and 20K profiling galleries by replicating the cached global embeddings and SGPA key-value features from the original 452-image RSITMD gallery. As shown in Table XX(b), at 20K, two-stage SGPA reduces latency from 6.619 ± 0.007 to 0.313 ± 0.001 ms/query, feature-cache memory from 3945.3 to 43.0 MB, and online memory increment from 2040.6 to 20.0 MB. We further use an HNSW index for the first stage. As shown in Table XX(c), HNSW achieves 100% Recall@20 relative to exact retrieval on both 5K and 20K galleries, with search latencies of 0.082 and 0.092 ms/query and index-memory costs of 11.08 and 44.33 MB, respectively.

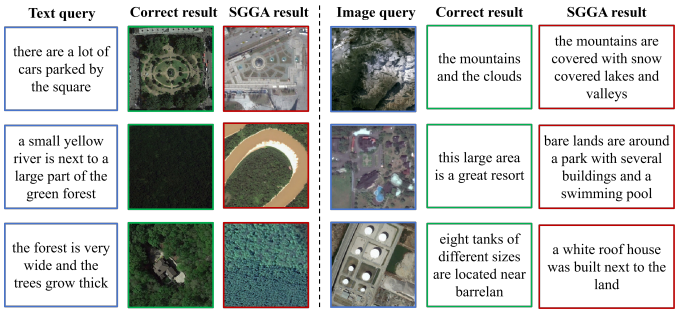


Fig. 10. Representative failure cases of SGGA on the RSITMD dataset. The left panel shows Text-to-Image retrieval and the right panel shows Image-to-Text retrieval. For each query, green is the correct result, while red is the incorrect result retrieved by SGGA.

VI. FAILURE CASE AND LIMITATION ANALYSIS

To further examine the limitations of SGGA, we present representative failure cases from both retrieval directions in Fig. 10, comparing the correct result with the incorrect result retrieved by SGGA. These cases show that SGGA may still confuse samples with highly similar global semantics while failing to distinguish subtle differences in local objects, spatial relationships, or scene composition. For example, scenes involving forests, rivers, resorts, and storage tanks can be incorrectly matched to visually or semantically similar alternatives. Although these predictions share partial semantic consistency with the queries, they differ from the correct results in fine-grained content. This indicates that SGGA remains less reliable when retrieval depends on subtle local structures, precise spatial relations, or small-scale objects.

VII. CONCLUSION

In this paper, we propose SGGA, a semantic-guided geometry-aware alignment framework for remote sensing image–text retrieval. Centered on SGPA for query-conditioned visual representation construction, SGGA further incorporates BNDA and SADR to regulate local neighborhood structure and global directional discrepancy in the shared embedding space. Through the joint optimization of these components, SGGA improves both the semantic relevance of visual representations and the local–global geometric consistency across modalities. During inference, a dual-query retrieval strategy

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further improves similarity estimation by combining complementary similarity scores from the original and gap-aware queries. Extensive experiments on RSITMD, RSICD, and UCM-Captions demonstrate the strong retrieval performance of SGGA, while ablation and quantitative geometry analyses verify the complementary roles of its components.

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